

Functional&Performance Testing Template

Model Performance Test

Date	02 October 2026
Team ID	6ab7f3e4b8bf817b7f724e1d
Project Name	Fitbuddy
Maximum Marks	

Test Scenarios&Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., fitness goal, feedback text)	Enter valid text (e.g., "muscle gain") and invalid/empty strings in input fields.	Valid inputs accepted, errors for invalid inputs.	FastAPI successfully validates string inputs and rejects empty feedback fields.	Pass
FT-02	Number Input Validation (e.g., age, weight)	Enter numbers within and outside the valid range.	Accepts valid values, shows error for out-of-range.	System accepts valid metrics (e.g., 75kg) and throws a 422 Unprocessable Entity error for negative numbers.	Pass
FT-03	Content Generation (e.g., 7-day workout plan)	Provide complete user inputs (age, weight, goal, intensity) and trigger generation.	Correct content is generated based on input.	Gemini AI returns a fully populated, goal-specific 7-day schedule in JSON format.	Pass
FT-04	API Connection Check	Check if API key is	API responds	Authenticati	Pass

		correct and Gemini model responds.	successfully.	on with Google GenAI SDK is successful, and endpoints return a 200 OK status.	
PT-01	Response Time Test	Use a timer to check content generation time for nutrition tips and workout plans.	Should be under 3 seconds.	Gemini Flash tip generation takes ~0.8s; Gemini Pro plan generation takes ~2.5s.	Pass
PT-02	API Speed Test	Send multiple API calls to the FastAPI backend at the same time.	API should not slow down.	Asynchronous FastAPI routing maintains stable response times under concurrent requests.	Pass
PT-03	Data Payload Load Test (e.g., large feedback text)	Submit extremely large text payloads during the feedback loop to check processing.	Should work smoothly without crashing.	The system safely truncates/processes large inputs without crashing the SQLite database or backend.	Pass